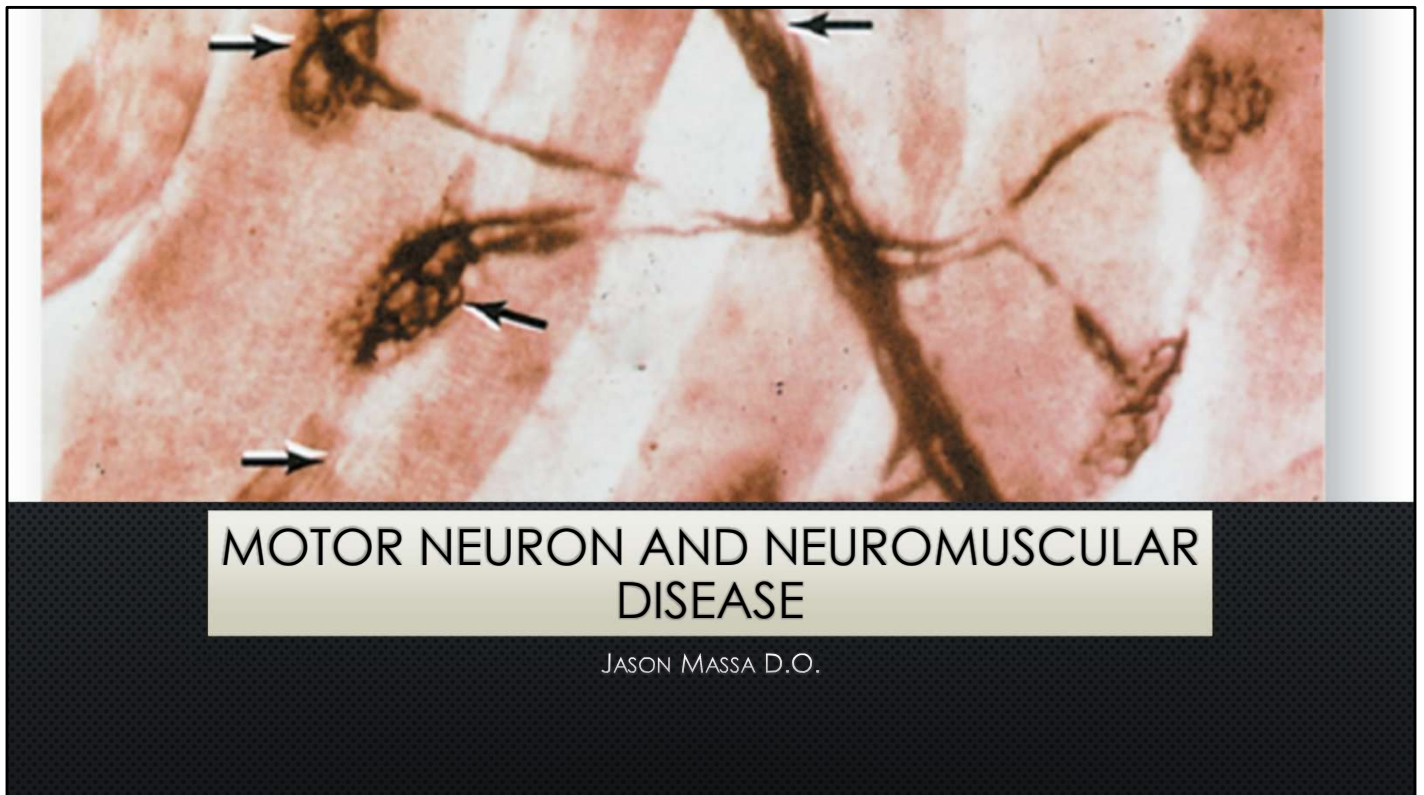




EM Picture of the neuromuscular endplates innervating muscle fibers, the location for this disease process.

OBJECTIVES

- RECOGNIZE COMMON PRESENTING COMPLAINTS IN PATIENTS WITH MOTOR NEURON AND NEUROMUSCULAR JUNCTION DISORDERS
- DESCRIBE COMMON CLINICAL FEATURES OF MYASTHENIA GRAVIS
- UNDERSTAND PHARMACOLOGIC TESTING FOR THE DIAGNOSIS OF MYASTHENIA GRAVIS
- DESCRIBE THE TREATMENT OPTIONS AVAILABLE FOR MYASTHENIA GRAVIS
- RECOGNIZE COMMON PRESENTING COMPLAINTS OF ALS
- UNDERSTAND PATHOPHYSIOLOGY OF ALS
- LEARN COMMON GENETIC FORMS OF ALS
- KNOW THE ROLE OF PHARMACOLOGIC TREATMENT IN ALS
- RECOGNIZE ALS MIMICS

MYASTHENIA GRAVIS AND RELATED DISEASE

EPIDEMIOLOGY

Prevalence from 43 to 84 million persons

Annual incidence rate 1 per 300,000

Any age of onset but onset rare in first decade

Peak age between 20-30 years in woman and 50-60 years in men

Under age 40, females 2:1 men

Later in life, men 3:2 women

Of patients with Thymoma, majority older (50-60 years) and males predominate

Up to 30% may have (+) FHx of autoimmune dz.

The disease may begin at any age, but onset in the first decade is relatively rare (only 10 percent of cases begin in children younger than 10 years of age).

COURSE AND PROGNOSIS

Course is variable. Tends to be steadily progressive.

Risk of death peaks in year one and after years 4-7 after onset, mainly from pneumonia and aspiration.

First year mortality now less than 5% (formerly 30%)

Conversion from ocular to generalized MG: Increasing duration of purely ocular MG associated with ↓ risk of generalized MG.

In MG patients presenting with ocular symptoms, weakness may remain restricted to ocular muscles in 15-20%. However, majority (90%) progress to generalized MG within first 12 months.

Spontaneous remission in first 1-2 years in 10% to 15% of cases, therefore slowly taper off immunosuppressive meds once disease is under control.

The *course of the illness* is extremely variable. Rapid spread from one muscle group to another occurs in some, but in others the disease remains unchanged for years before progressing or there is no progression. Remissions may take place without explanation, usually in the first years of illness, but these happen in less than half the cases and seldom last longer than a month or two.

An aspect of interest is the timing and frequency of conversion from ocular and restricted oropharyngeal patterns of weakness to more widespread involvement including the diaphragm. [Bever and coworkers](#) have confirmed the general impression that an increasing duration of purely ocular myasthenia is associated with a decreasing risk of late generalization of weakness. In a retrospective study of 108 patients, these authors found that only 15 percent of the observed instances of generalization occurred more than 2 years after isolated ocular manifestations.

ETIOLOGY AND PATHOGENESIS

Failure of effective neuromuscular transmission-postsynaptic membrane

Blocked transmission at endplates results in reduced contractile power of the muscle.

Deficiency first in muscles that are continuously active and fewest AChRs per muscle (ocular)

Fatigue: result of normal decline of amount of Ach released with each successive impulse.

CLINICAL MANIFESTATIONS

Fluctuating weakness of voluntary (skeletal) muscles primarily those innervated by the motor nuclei of the brainstem, i.e., ocular, masticatory, facial.

Weakening during continued activity, quick restoration of power with rest, dramatic improvement in strength following anticholinesterase drugs. Special vulnerability of NMJ in certain muscles

Initial presentation: Eye lid, eye movement > face, jaws, throat, neck > limb and respiratory

Ocular palsies or ptosis in ~50% and eventually ~90% involvement

Diplopia inconsistent with III, IV, VI innervation

Spreads insidiously to limbs and axial muscles

highly characteristic clinical appearance. Usually the eyelids and the muscles of eye movement, and somewhat less often, of the face, jaws, throat, and neck, are the first to be affected.

Specifically, weakness of the levator palpebrae or extraocular muscles is the initial manifestation of the disease in about half the cases, and these muscles are involved eventually in more than 90 percent of cases.

Diplopia is common in myasthenia, but it does not correspond to the innervatory pattern of a nerve; spreads insidiously from the cranial to the limb and axial muscles, but there are instances of fairly rapid development, sometimes initiated by an infection, usually respiratory.

CLINICAL MANIFESTATIONS

Tendency to get worse as the day progresses

Smile turns to snarl

Chewing becomes difficult leading to weight loss

Voice fades and becomes nasal after sustained conversation

Shoulder girdle weakness can manifest as difficulty doing hair, makeup. Camptocormia (flexed head)

CLINICAL MANIFESTATIONS

Muscles of facial expression, mastication, swallowing, speech affect 80% of patients at some time and 5-10% are the initial muscles.

Less frequent: neck flexors and extensors, shoulder girdle, hip flexors

Of trunk muscles, erector spinae more often affected.

Most advanced cases: all muscles including diaphragmatic, abdominal and intercostal, external sphincters of bladder and bowel.

Minimal atrophy

Deep tendon reflexes seldom altered Aching of weakened muscles Tongue atrophy rare (other than MuSK syndrome- muscle specific tyrosine kinase disease)- we will discuss later.

FUNCTIONAL HISTORY

Arising from chair

Climbing stairs

Eating/jaw fatigue

Head/neck fatigue

Fixing hair, makeup, curling iron,

Lipstick (unable to purse and roll lips)

Unable to control flatus

OCULAR SIGNS

Sustained upgaze 30 seconds worsens ptosis

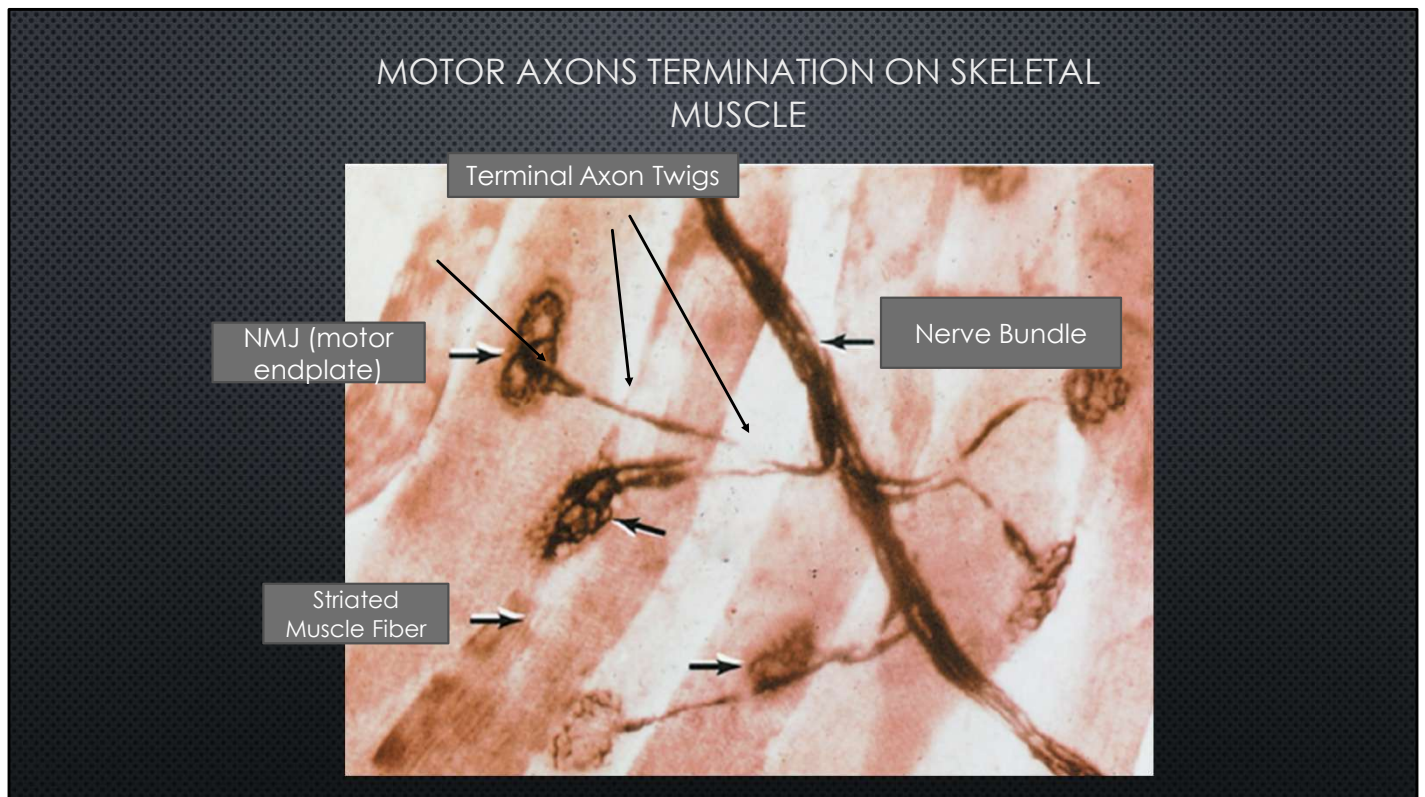
Cogan's Sign (twitch upper lid when moving eyes from downward position to neutral)

Repeated ocular versions or optokinetic stimulus discloses paresis

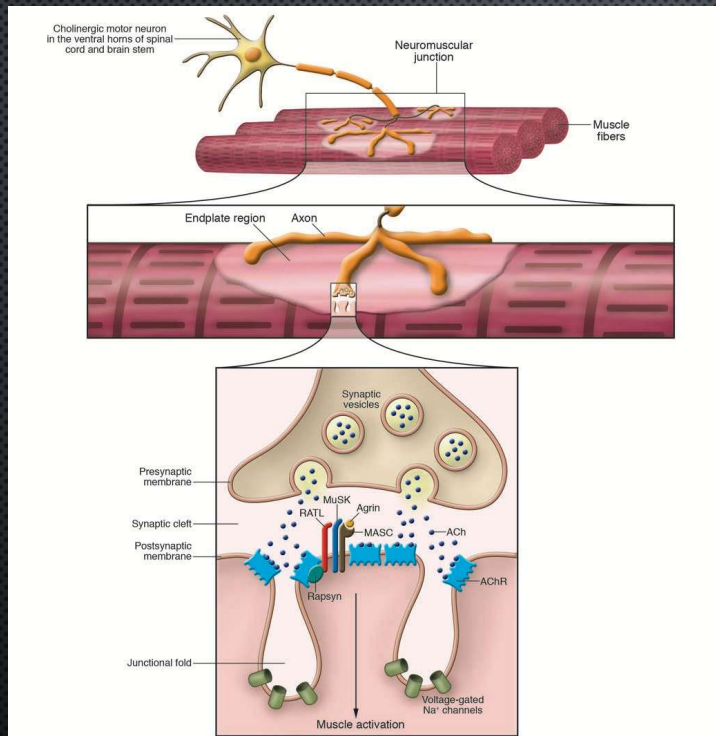
Subtle ptosis of eye when elevating more symptomatic eyelid

Worse with bright sun, better with cold (pack)

sustained upgaze for 30 or more seconds will usually induce or exaggerate ptosis and may uncover myasthenic ocular motor weakness. [Cogan](#) described a twitching of the upper eyelid that appears a moment after the patient moves the eyes from a downward to the primary position ("lid-twitch" sign). Or, after sustained upward gaze, one or more twitches may be observed upon closure of the eyelids or during horizontal movements of the eyes. Repeated ocular versions when tracking a target or by an optokinetic stimulus may disclose progressive paresis of the muscles that carry out these movements. Bright sunlight is said to aggravate the ocular signs and cold to improve them. The application of an ice pack over the eye often relieves the ptosis for a brief period.

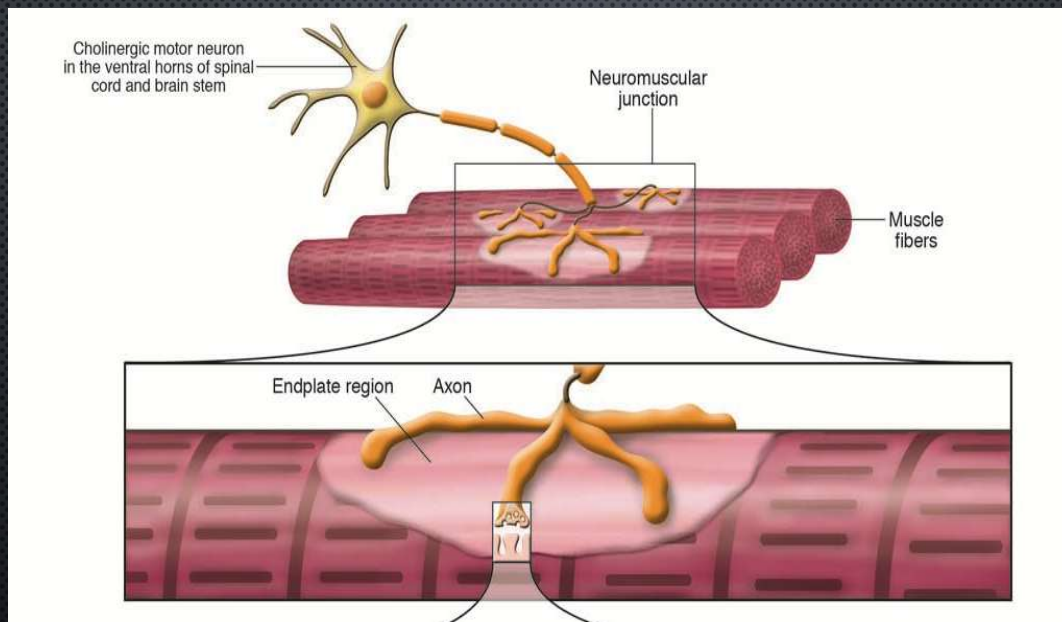


Before it terminates in a skeletal muscle, each motor axon bundled in the nerve forms many branches, each of which forms a synapse with a muscle fiber.(a) Silver staining can reveal the nerve bundle (NB), the terminal axonal twigs, and the motor end plates (MEP, also called neuromuscular junctions or NMJ) on striated muscle fibers (S). X1200.(b) An SEM shows the branching ends of a motor axon, each covered by an extension of the last Schwann cell and expanded terminally as an MEP embedded in a groove in the external lamina of the muscle fiber.(c) Diagram of enclosed portion of the SEM indicating key features of a typical MEP: synaptic vesicles of acetylcholine (ACh), a synaptic cleft, and a postsynaptic membrane. This membrane, the sarcolemma, is highly folded to increase the number of ACh receptors at the MEP. Receptor binding initiates muscle fiber depolarization, which is carried to the deeper myofibrils by the T tubules.



Conti-Fine B. J Clin. Invest. 116: 1243-2854 (2006)
doi:10.1172/JCI29894.

Let's take a closer look at the neuromuscular junction- next slide.



Conti-Fine B. J Clin. Invest. 116: 1243-2854 (2006)
doi:10.1172/JCI29894.

Structure of the NMJ. As it enters the muscle and approaches its target fibers, each α -motor neuron axon divides into branches that innervate many individual muscle fibers. Each branch loses its myelin sheath and further subdivides into many presynaptic boutons, which contain ACh- loaded synaptic vesicles and face the surface of the muscle fiber at the NMJ. The synaptic bouton and the muscle surface are separated by the synaptic cleft, which contains AChE and proteins and proteoglycans involved in stabilizing the NMJ structure.

IMMUNOPATHOGENESIS of MYASTHENIA GRAVIS

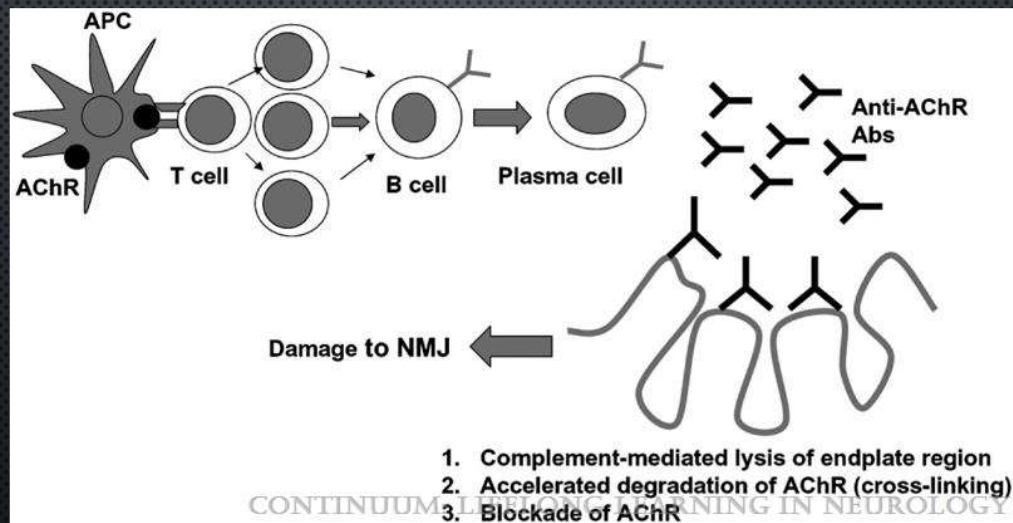


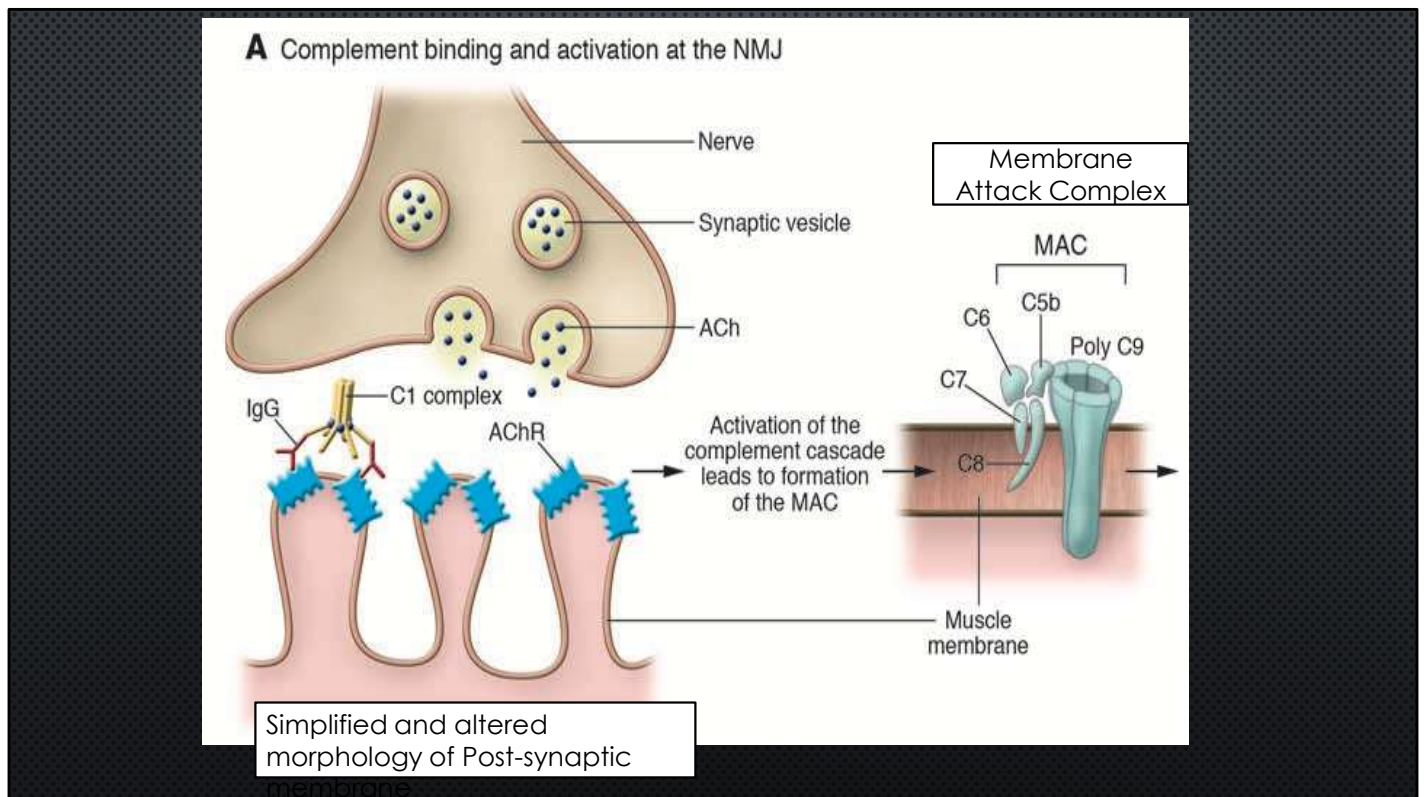
FIGURE 2-1

The immunopathogenesis of myasthenia gravis involves the production of high-affinity anti-acetylcholine receptor (AChR) antibodies whose synthesis is modulated by and dependent on AChR-specific T cells that are activated after presentation of AChR peptides by antigen-presenting cells (APC).

Autoantibodies reduce the numbers of functional AChRs mainly by complement-mediated lysis of the postsynaptic membrane and by cross-linking of AChRs, causing enhanced endocytosis and degradation.

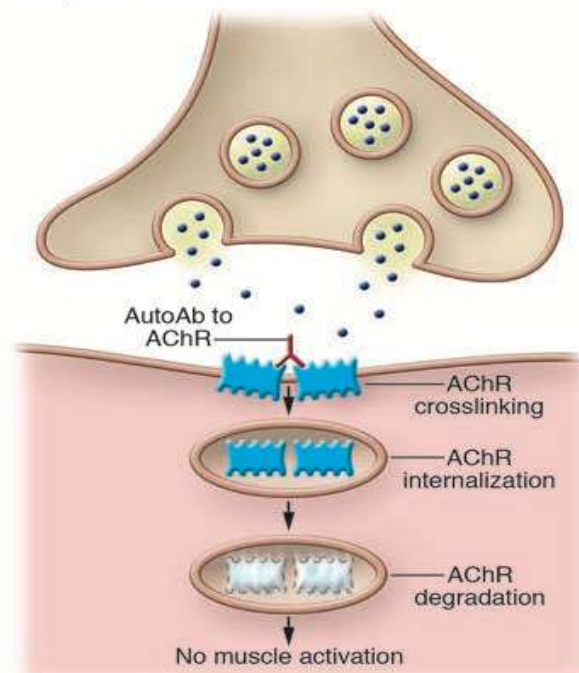
Direct blockade of the acetylcholine-binding site on the AChR is a less frequent mechanism.

Abs = antibodies; NMJ = neuromuscular junction



Effector mechanisms of anti-AChR Abs. **(A)** Ab binding to the AChR activates the complement cascade, resulting in the formation of membrane attack complex (MAC) and localized destruction of the postsynaptic NMJ membrane. This ultimately leads to a simplified, altered morphology of the postsynaptic membrane of the NMJ of MG patients, which lacks the normal deep folds and has a relatively flat surface.

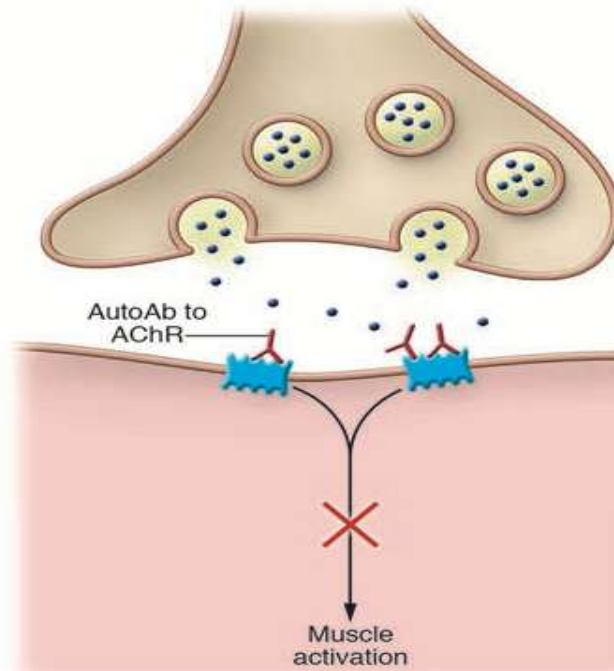
B Antigenic modulation



Cross-link
AChR
molecules on
the NMJ post-
synaptic
membrane
cause
endocytosis
and
degradation

(B) Abs cross-link AChR molecules on the NMJ postsynaptic membrane, causing endocytosis of the cross-linked AChR molecules and their degradation (antigenic modulation). This ultimately leads to a reduced number of AChR molecules on the postsynaptic membrane.

C Functional AChR block



(C) Ab binding the ACh-binding sites of the AChR causes functional block of the AChR by interfering with binding of ACh released at the NMJ. This results in failure of neuromuscular transmission.

EXAMINATION

Normal

No or minimal atrophy

DTR's (deep tendon reflexes)
normal

Normal sensory exam

Patient's don't complain about neuromuscular junction problems, they complain about weakness. Step one is to determine that the NMJ is the site of pathology. A detailed neurologic exam will help to exclude any upper motor neuron or motor neuron disease.



30 SECOND
UPWARD
GAZE



CLINICAL MANIFESTATIONS

SNARL FOR A SMILE

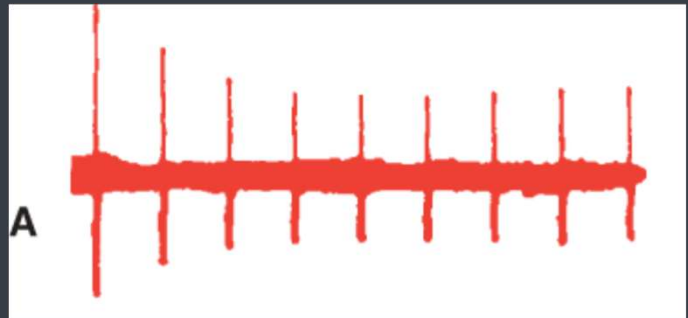


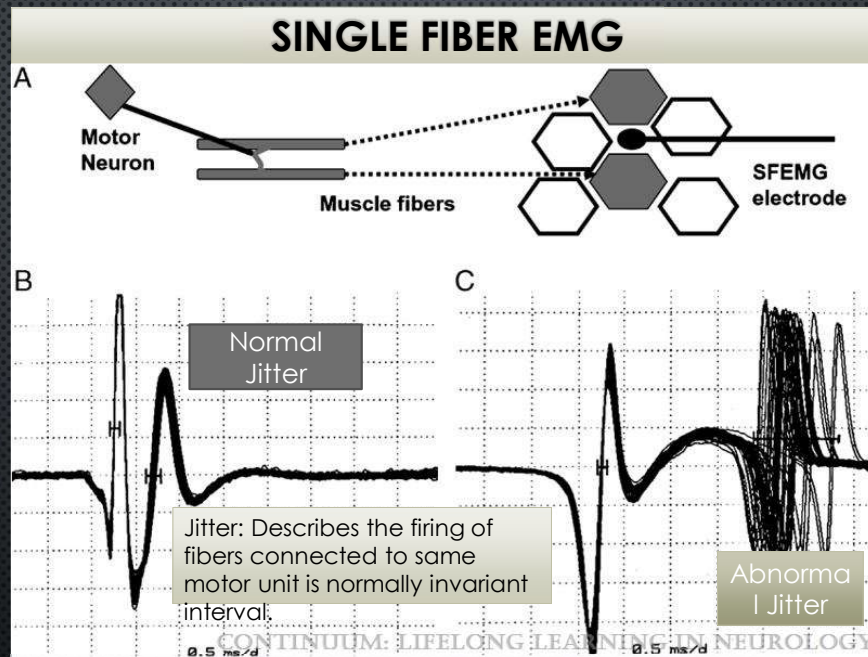
OTHER EXAMINATION TECHNIQUES:

- ICE-PACK TEST – WHEN SUSPECTING MG AS A CAUSE OF PTOSIS, YOU CAN PUT AN ICE PACK OVER THE LID OF THE AFFECTED EYE FOR ABOUT 2 MINUTES (COVERED TO PROTECT THE EYE.) IF THE PTOSIS IMPROVES, THIS IS SUGGESTIVE OF NMJ SOURCE OF WEAKNESS AS NMJ TRANSMISSION IS IMPROVED WITH COOL TEMPERATURE.
- RAPID ALTERNATING MOVEMENTS OF ANY AFFECTED EXTREMITY – CAN DO WING-FLAPPING MOVEMENTS FOR SHOULDER WEAKNESS OR FOOT TAPPING FOR ANKLE WEAKNESS. CHECK FOR FATIGUING WEAKNESS AND BE SURE TO MAKE BEFORE AND AFTER COMPARISON TO THE CONTRALATERAL SIDE (IF CONTRALATERAL SIDE UNAFFECTED.)

ELECTRODIAGNOSTIC TESTING

- NERVE CONDUCTION STUDY
 - REPETITIVE STIMULATION
- ELECTROMYOGRAPHY
 - SINGLE FIBER EMG
 - A **DECREMENTAL** response with REPETITIVE NERVE STIMULATION. NOTICE THE LOWERING OF THE AMPLITUDE OF THIS COMPOUND MOTOR ACTION POTENTIAL (CMAP). THE CMAP IS THE SUMMATED VOLTAGE RESPONSE OF MANY MOTOR FIBERS LOCATED IN CLOSE GEOGRAPHIC PROXIMITY.





MERIGGIOLI, MATTHEW N.

CONTINUUM: LIFELONG LEARNING IN NEUROLOGY, 15(1, MYASTHENIC DISORDERS AND ALS):35-62, FEBRUARY 2009. DOI: 10.1212/01.CON.0000300007.56974.b7

Wolters Kluwer

Copyright © 2016 American Academy of Neurology. Published by Lippincott Williams & Wilkins.

25

FIGURE 2-4 A, In single fiber EMG (SFEMG) testing, the recording electrode is positioned to record from two or more time-locked potentials.

Jitter is measured as the variation in the time interval between the two action potentials in the pair. B, An example of normal jitter; 40 consecutive discharges are superimposed. C, Abnormal jitter; 50 consecutive discharges are superimposed.

Concentric needle records AP from individual muscle fibers. Variation in latency from nerve action to muscle action is the neuromuscular jitter and is produced by fluctuations in the time it takes for the EPP at the NMJ to reach the threshold for action-potential generation. Performed during voluntary activation of the muscle. Measured as the variation in the time interval between the two action potentials in the pair (interpotential interval). An increase in magnitude of this jitter is the most sensitive electrophysiologic sign of a defect in NM transmission. When severe, some nerve impulses fail to elicit Aps and SFEMG demonstrate an intermittent absence or one or more single muscle fiber action potentials on consecutive firings. This is called impulse blocking and represents NM transmission failure at the involved endplate. Sensitivity ranges from 82 to 99% and highest when attaining up to 3 muscles tested. Jitter may be increased in primary nerve or muscle disease, so not specific, but very sensitive. NCV and conventional EMG done in all cases to rule out nerve and muscle disease.

NEOSTIGMINE TEST

Anticholinesterase inhibitor, rapid action,
improves weakness

Preload with atropine (avoid salivation, sweating,
bowel cramps, pulmonary secretions, diarrhea)
followed by IM 1.5 mg.

Improvement in 10-15 minutes, peak at 20
minutes, lasts up to 1 hour.

Consider performance with control.

(-) test does not r/o MG

SEROLOGIC TESTING

AChR binding Ab present in ~80% with generalized MG; but only 55% of ocular MG.

Some initially seronegative, may convert several months after disease onset.

About 4% with negative AChR binding Ab demonstrate modulating Ab and 1% demonstrate blocking Ab.

Anti-Striated Muscle Ab or anti-striational Ab found in 30% of patients with adult-onset MG.

Useful for serologic marker in younger patient with thymoma.

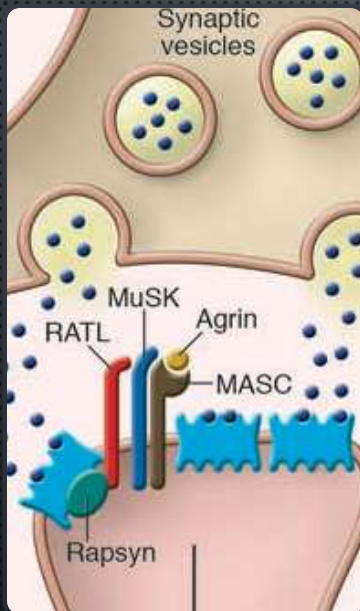
ACETYLCHOLINE ANTIBODIES

Persistently negative AChR Ab tests more often with ocular MG

Patients with thymoma and severe generalized MG practically always sero (+)

Ab titers often remain elevated during clinical remissions

About 1/3rd of SN-MG (seronegative MG) are MuSK antibody positive



ANTI-MUSK MUSCLE-SPECIFIC TYROSINE KINASE

- RESPONSIBLE FOR MAINTAINING STRUCTURE OF NM AND CLUSTERING OF THE AChR'S ON THE POST-SYNAPTIC MEMBRANE.

Instances of "seronegative" disease are sometimes due to antibody production against unusual muscle epitopes that are located on or near the acetylcholine receptor; their detection requires a special panel of tests. However, the majority of such cases have been ascribed to IgG antibodies directed against an intracellular *muscle-specific kinase* (MuSK). This enzyme plays a role in supporting the normal structure of the postsynaptic membrane and in the arrangement of AChR but its main function may be in developmental synaptic differentiation. [Scuderi and colleagues](#) and others have proposed that patients with MuSK antibody, mostly women, have a special clinical syndrome of prominent oculobulbar weakness, often with severe disease and respiratory crises (see also [Evoli et al](#)). Others have reported a different pattern of mainly neck and proximal weakness that simulates a typical myopathy. Many of these patients are inadequately responsive to anticholinesterase treatment. Also of interest, but not currently used in routine diagnosis, is the presence of *antibodies directed against striated muscle* in almost half of myasthenic patients and an even higher incidence (stated to be 85 percent) in patients who also have a thymoma.

ANTI-MUSK MUSCLE- SPECIFIC TYROSINE KINASE

- APPROXIMATELY 50% OF PATIENTS WITH GENERALIZED, AChR-AB NEGATIVE MG HAVE MUSK-MG.
- REPRESENTS ~5-8% OF ALL PATIENTS WITH MG
- APPROXIMATELY 85% FEMALE, 4TH DECADE, USUALLY MORE SEVERE IN ONSET WITH FREQUENT RESPIRATORY CRISIS.
- PROMINENT OCULAR/BULBAR AND AXIAL SYMPTOMS.
- PLASMA EXCHANGE PRODUCED IMPROVEMENT IN 93% AND LONG-TERM OUTCOME COMPARABLE TO AChR-AB-POSITIVE MG.
- LESS FLUCTUATION, LACK OF RESPONSE TO AChE INHIBITORS AND THYMECTOMY. MANY CORTICOSTEROID DEPENDENT

MEDICATION TREATMENT

THREE MAIN GROUPS OF MEDICATIONS

1. CHOLINESTERASE INHIBITORS
2. IMMUNOMODULATORY AGENTS
3. BIOLOGIC AGENTS

CHOLINESTERASE INHIBITORS

- FOR SYMPTOMATIC TREATMENT
- FIRST LINE TX FOR ALL PATIENTS WITH MG
- PYRIDOSTIGMINE IS MOST COMMON:
 - DOSE 30-60 MG EVERY 4-6 HOURS
 - MAXIMAL DOSE <300 MG/DAY
 - BEST TAKEN 30 MINUTES PRIOR TO MEAL (TO IMPROVE SWALLOWING)
 - CHOLINERGIC SE: N/V, ABDOMINAL CRAMPS, DIARRHEA, INCREASED ORAL AND BRONCHIAL SECRETIONS, BRADYCARDIA, RARELY PSYCHOSIS

IMMUNOMODULATORY AGENTS

- CORTICOSTEROIDS
- AZATHIOPRINE
- CYCLOSPORIN
- TACROLIMUS
- MYCOPHENOLATE MOFETIL
- METHOTREXATE
- CYCLOPHOSPHAMIDE

CORTICOSTEROIDS

- ***CONSIDERED THE MOST EFFICACIOUS ORAL IMMUNOTX.**

- **MILD:** PREDNISONE AT LOW DOSE (10-20 MG/DAY) IN OUT-PATIENT SETTING, SLOWLY INCREASE BY 5 MG EVERY 5-7 DAYS UNTIL SYMPTOMS IMPROVE UP TO MAXIMUM OF 50 MG A DAY. AFTER STABLE FOR 4 WEEKS, TAPER SLOWLY TO LOWEST AMOUNT TOLERABLE.

- **MODERATE-SEVERE:** START HIGH-DOSE (0.75-1 MG/KG/D) FOR INPATIENT. TAPER BY 5-10 MG PER MONTH UNTIL PREDNISONE DOSEAGE IS 20 MG/DAY AND THEN BY 5 MG/MONTH UNTIL THEY ARE ON 10 MG/DAILY, THEN 1-2.5 MG A MONTH.

Corticosteroid Side Effects

Side Effect	Prevention
Sodium/fluid retention	Sodium-restricted diet
Obesity	Low-calorie, low-fat diet; exercise
Potassium loss	Supplementation as needed
Hypertension	Monthly checks, treatment
Impaired glucose tolerance	Fasting blood glucose monitored, treated
Osteoporosis	Calcium/vitamin D supplementation, bone density evaluation, bisphosphonates, female hormone replacement
Cataracts/glaucoma	At least yearly ophthalmic evaluation
Steroid myopathy	Exercise, high-protein diet
Suppression of growth (children)	Minimum effective dose used
Peptic ulcer	Treatment if symptoms occur

CONTINUUM: LIFELONG LEARNING IN NEUROLOGY

NONSTEROID IMMUNOMODULATING AGENTS

- AZATHIOPRINE:

- DELAYED ONSET FOR MONTHS
- EXCLUDE THIOMETHYL PURINE TRANSFERASE DEFICIENCY BEFORE STARTING THERAPY.
- SUPPORTED BY EXPERT CENSUS AND SOME RCT'S

- CYCLOSPORIN:

- EARLIER ONSET OF ACTION THAN AZA BUT LESS WELL TOLERATED.
- HTN AND NEPHROTOXICITY
- EVIDENCE IN SOME RCT'S

- MYCOPHENOLATE AND TACROLIMUS: NO RCT EVIDENCE, BUT WIDELY USED AND RECOMMENDED IN GUIDELINES.

BIOLOGIC AGENTS

- RITUXIMAB:

- MONOCLONAL AB AGAINST CD20+ B CELLS
- MUSK MG RESPONDS WELL/LONG REMISSIONS

- ECULIZUMAB

- HUMANIZED MURINE MONOCLONAL AB
- IN US, APPROVED FOR GENERALIZED ACHR MG; IN EUROPE FOR REFRACTORY GENERALIZED ACHR MG.
- BINDS TO COMPLEMENT PROTEIN C5 TO INHIBIT CLEAVAGE OF C5 TO C5A AND C5B, BLOCKING TERMINAL COMPLEMENT ACTIVATION (CALLED MEMBRANE ATTACK COMPLEX).

IVIG AND PLASMA EXCHANGE

- BOTH PROBABLY EQUALLY EFFICACIOUS.
- USED FOR SHORT-TERM TX FOR LIFE-THREATENING SIGNS, (RESPIRATORY INSUFFICIENCY OR DYSPHAGIA); IN PREPARATION FOR SURGERY IN PATIENTS WITH SIGNIFICANT BULBAR DYSFUNCTION; WHEN A RAPID RESPONSE TO TREATMENT IS NEEDED; WHEN OTHER TREATMENTS ARE INSUFFICIENTLY EFFECTIVE; AND PRIOR TO BEGINNING CORTICOSTEROIDS IF DEEMED NECESSARY TO PREVENT OR MINIMIZE EXACERBATIONS.
- THE EFFICACY OF IVIG IS LESS CERTAIN IN Milder MG OR IN OCULAR MG.
- PLEX MAY BE MORE EFFECTIVE THAN IVIG IN MUSK-MG.
- THE USE OF IVIG AS MAINTENANCE THERAPY CAN BE CONSIDERED FOR PATIENTS WITH REFRACTORY MG OR FOR THOSE IN WHOM IS AGENTS ARE RELATIVELY CONTRAINDICATED.

THYMECTOMY

•NON-CONTRAST CT SHOULD BE PERFORMED IN PATIENTS WITH AChR BINDING AB TO EXCLUDE THYMOMA. CHEST MRI DOES NOT IMPROVE SENSITIVITY.

•IMPORTANT!

•THYMECTOMY HAS NOW BEEN SHOWN TO IMPROVE OUTCOMES IN PATIENTS 18 – 65 YEARS OLD WITH NONTHYMOMATOUS MYASTHENIA GRAVIS

•THAT'S NONTHYMOMATOUS MG!

•RCT: THYMECTOMY + ALTERNATE-DAY PREDNISONE V. ALT-DAY PREDNISONE ALONE

•N:= 126; 2006-2012, AGED 18-65; 36 SITES; 3 YEAR END POINT

•RESULTS: LOWER QUANTITATIVE MG SCORE, LOWER ALTERNATE-DAY PREDNISONE DOSE, FEWER HOSPITALIZATIONS, REDUCED REQUIREMENT FOR ADDITIONAL IMMUNOSUPPRESSION IN FORM OF AZATHIOPRINE

The NEW ENGLAND JOURNAL of MEDICINE

ESTABLISHED IN 1812

AUGUST 11, 2016

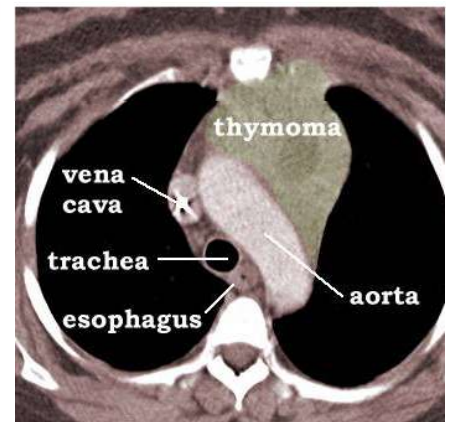
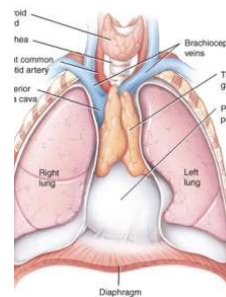
VOL. 375 NO. 6

Randomized Trial of Thymectomy in Myasthenia Gravis

G.J. Wolfe, H.J. Kaminski, I.B. Aban, G. Minisman, H.-C. Kuo, A. Marx, P. Ströbel, C. Mazza, J. Oger, J.G. Cea, J.M. Heckmann, A. Evoli, W. Nix, E. Cufaloni, G. Antonini, R. Witoonpanich, J.O. King, S.R. Beydoun, C.H. Chalk, A.C. Barboi, A.A. Amato, A.I. Shaibani, B. Katiqi, B.R.F. Lecky, C. Buckley, A. Vincent, E. Dias-Tosta, H. Yoshikawa, M. Waddington-Cruz, M.T. Pulley, M.H. Rivner, A. Kostera-Pruszczyk, R.M. Pascuzzi, C.E. Jackson, G.S. Garcia Ramos, J.J.G.M. Verschuuren, J.M. Massey, J.T. Kissel, L.C. Werneck, M. Benatar, R.J. Barohn, R. Tandan, T. Mozaffar, R. Conwit, J. Odenkirchen, J.R. Sonett, A. Jaretski, III, J. Newsom-Davis, and G.R. Cutter, for the MGTX Study Group[®]

BACKGROUND

Thymectomy has been a mainstay of treatment for myasthenia gravis, but there has been no conclusive evidence of its benefit. This study compared thymectomy plus prednisone with prednisone alone.



THYMECTOMY IN NON- THYMOMATOUS MG

- OPTION TO POTENTIALLY AVOID OR MINIMIZE DOSE OR DURATION OF IMMUNOTHERAPY
- OR FOR FAILED RESPONSE TO INITIAL IMMUNOTHERAPY OR INTOLERABLE SIDE-EFFECTS
- PERFORMED WHEN PATIENT IS STABLE AND DEEMED SAFE TO UNDERGO THE PROCEDURE.

Sanders DB, Wolfe GI, Benatar M, Evoli A, Gilhus NE, Illa I, Kuntz N, Massey JM, Melms A, Murai H, Nicolle M, Palace J, Richman DP, Verschuuren J, Narayanaswami P. International consensus guidance for management of myasthenia gravis: Executive summary. *Neurology*. 2016 Jul 26;87(4):419-25.

THYMECTOMY IN MG WITH THYMOMA

- WITH RARE EXCEPTION, ALL PATIENTS SHOULD UNDERGO THYMECTOMY (ENTIRE THYMUS PLUS TUMOR)
- IN ELDERLY OR MULTIMORBID PATIENTS, PALLIATIVE RADIATION THERAPY CAN BE CONSIDERED
- NO EVIDENCE FOR MU_{SK}, LRP4 OR AGRIN ANTIBODIES
- ENDOSCOPIC AND ROBOTIC APPROACHES IN EXPERIENCED HANDS OFFER SIMILAR RESULTS.
- MAY BE CONSIDERED WITH GENERALIZED MG W/O DETECTABLE AChR AB IF THEY FAIL TO RESPOND ADEQUATELY TO IS Tx OR TO AVOID/MINIMIZE INTOLERABLE ADVERSE EFFECTS FROM IS Tx.

Sanders DB, Wolfe GI, Benatar M, Evoli A, Gilhus NE, Illa I, Kuntz N, Massey JM, Melms A, Murai H, Nicolle M, Palace J, Richman DP, Verschuuren J, Narayanaswami P. International consensus guidance for management of myasthenia gravis: Executive summary. *Neurology*. 2016 Jul 26;87(4):419-25.

MYASTHENIC CRISIS

- SEVERE EXACERBATION OF MG THAT MAY ENDANGER LIFE DUE TO RESPIRATORY MUSCLE WEAKNESS AND BULBAR DYSFUNCTION (RISK OF ASPIRATION).
- CAN OCCUR IN 15-20% OF CASES USUALLY IN FIRST 1-2 YEARS.
- ICU-CLOSELY MONITOR
- Tx: PLEX OR IVIG FOLLOWED BY HIGH-DOSE CORTICOSTEROIDS

CONDITIONS AND MEDICATIONS WHICH MAY EXACERBATE OR INDUCE MYASTHENIC CRISIS:

- ◆ Magnesium and magnesium-containing medications (eg, laxatives, antacids)
- ◆ Botulinum toxin
- ◆ Interferon alfa
- ◆ D-Penicillamine
- ◆ Cardiovascular medications
 - ◇ Quinidine, quinine
 - ◇ Beta-blockers
 - ◇ Calcium channel blockers
- ◆ Anesthetics (eg, methoxyflurane)
- ◆ Neuromuscular blockers (eg, succinylcholine)
- ◆ Checkpoint inhibitors (eg, pembrolizumab)

Surgery

Pregnancy and Postpartum Period

Heat

Stress

Viral Infections

Bone Marrow Transplantation

Medications

◆ Antibiotics

- ◇ Aminoglycosides
- ◇ Fluoroquinolones
- ◇ Tetracyclines
- ◇ Sulfonamides
- ◇ Penicillins

DIFFERENTIAL DIAGNOSIS OF MYASTHENIA

Disease	Points of differentiation
Lambert Eaton Myasthenic Syndrome	Relative sparing ocular ms; hyporeflexia; dry mouth; positive Anti-VG Ca Channel Ab; low CMAP amp.
Congenital Myasthenic Sx	Onset infancy or childhood; no response to immunomodulation; abnormal Rep Stim.
Botulism	Rapid descending pattern of weakness; pupillary autonomic involvement, low amp CMAP
ALS/MND	Corticospinal or corticobulbar features; cramps, fasciculations; abnl. Single fiber EMG
Mitochondrial dz	Gradual onset, no fluctuation, often no diplopia, despite severe ophthalmoplegia
AIDM variants	No fluctuation of weakness; areflexia; acute onset; abnormal NCV and CSF
Thyroid ophthalmopathy	Proptosis
CNS Disorders causing cranial neuropathies	Sudden onset, altered LOC, impaired sensation, ataxia, diplopia w/in distribution of single nerve

LAMBERT- EATON SYNDROME

- LEMS IS A PARANEOPLASTIC SYNDROME. MOST COMMONLY SEEN WITH **SMALL CELL LUNG CARCINOMA** (60%)
- MUSCLE PATTERN: TRUNK, SHOULDER GIRDLE, PELVIC GIRDLE, LOWER EXTREMITIES MORE COMMON PRESENTATION. THINK FUNCTIONAL EXAMPLES.
- OFTEN PRESENT WITH LEG WEAKNESS AND PAIN
- TEMPORARY ↑ IN POWER AFTER 1ST FEW CONTRACTIONS
- REDUCED DTRs
- MILD AUTONOMIC DYSFUNCTION IS COMMON (AND DIFFERENTIATES FROM MG)-DRY MOUTH, ED, ORTHOSTATIC HYPOTENSION, CONSTIPATION)

LAMBERT- EATON SYNDROME

- ONSET IS SUBACUTE, COURSE VARIABLY PROGRESSIVE
- MALES FAR MORE OFTEN THAN FEMALES (5:1)
- LIKE ALL PARANEOPLASTIC SYNDROMES, MAY PRECEDE DISCOVERY OF TUMOR BY MONTHS- YEARS
- 1/3RD- NO TUMOR FOUND; SOME AUTOIMMUNE

LAMBERT- EATON SYNDROME

- POOR OR UNPREDICTABLE RESPONSE TO NEOSTIGMINE AND PYRIDOSTIGMINE
- FAST RATES OF NCV STIMULATION (50 Hz) OR AFTER STRONG VOLUNTARY CONTRACTION, INCREASED AMPLITUDE OF THE AP ("INVERSE MYASTHENIA")
- IGG ANTIBODIES CAUSE CROSS-LINKING OF VOLTAGE-GATED CALCIUM CHANNELS ON PRESYNAPTIC MOTOR NERVE TERMINAL, REDUCING NUMBER OF FUNCTIONAL CHANNELS
- REDUCES RELEASE OF ACH FROM NERVE TERMINALS (AKIN TO BOTULINUM TOXIN.
- POST-SYNAPTIC SURFACE AREA INCREASED
- NORMAL MUSCLE BIOPSY AND NORMAL THYMUS
- USUALLY, AGE > 50 YEARS

PART 2: MOTOR NEURON DISEASE

STOP
ALS



MOTOR NEURON DISEASE

- A SPECTRUM OF DISEASES CHARACTERIZED BY PROGRESSIVE MUSCULAR WEAKNESS
- DUE TO DEGENERATION OF MOTOR NEURONS WITHIN THE BRAIN, BRAINSTEM, AND SPINAL CORD
- THE MOST WELL KNOWN OF WHICH IS AMYOTROPHIC LATERAL SCLEROSIS (SPINAL FORM)

MOTOR NEURON SPECTRUM OF DISEASE

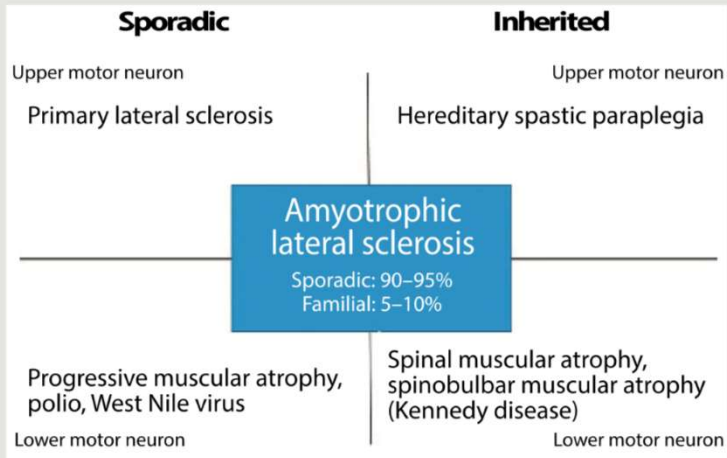


FIGURE 6-1 The spectrum of motor neuron diseases.

ALS EPIDEMIOLOGY

- MEAN AGE OF ONSET IS 55 YEARS, THOUGH THERE IS A WIDESPREAD AND CAN AFFECT THE EXTREMES OF AGE
- INCIDENCE 1.4 PER 100,000
- PREVALENCE 5 PER 100,000
- MALE:FEMALE - 1.7:1
- ETHNIC AND RACIAL DISPARITIES -> AFFECTS EUROPEAN DESCENDED INDIVIDUALS AT HIGHER RATE
- MEDIAN SURVIVAL IS 20 – 48 MONTHS
- **WHILE INCIDENCE AND PREVALENCE SEEM RELATIVELY LOW, GIVEN THE SHORT DURATION OF DISEASE, ALS IS NOT AS RARE AS IT SEEMS**
- **LIFETIME RISK IS ~ 1 IN 300, TO 1 IN 400**

ALS
PATHOLOGY
PATHOPHYSIOLOGY

- DEGENERATION OF UPPER MOTOR NEURONS AND ALPHA MOTOR NEURONS (LMN)
- "LATERAL SCLEROSIS" REFERS TO LATERAL CORTICOSPINAL TRACT DEGENERATION
- "AMYOTROPHY" MEANS WEAKNESS WITH ATROPHY

PATHOLOGICAL FINDINGS

- **BUNINA BODIES** - CYTOPLASMIC EOSINOPHILIC INTRACELLULAR INCLUSIONS ARE FOUND IN THE SPINAL CORD AND BRAINSTEM MOTOR NUCLEI.
- CYTOPLASMIC UBIQUITIN-POSITIVE **SKEINLIKE INCLUSIONS (TDP-43 PROTEIN INCLUSIONS)** FOUND IN ANTERIOR HORN CELLS, BUT ALSO FOUND IN FRONTAL, TEMPORAL, HIPPOCAMPUS, AND STRIATUM AS SEEN IN FRONTOTEMPORAL DEMENTIA.

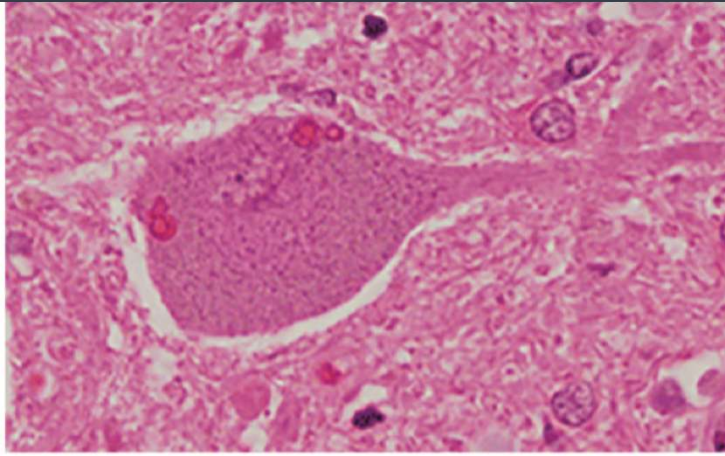
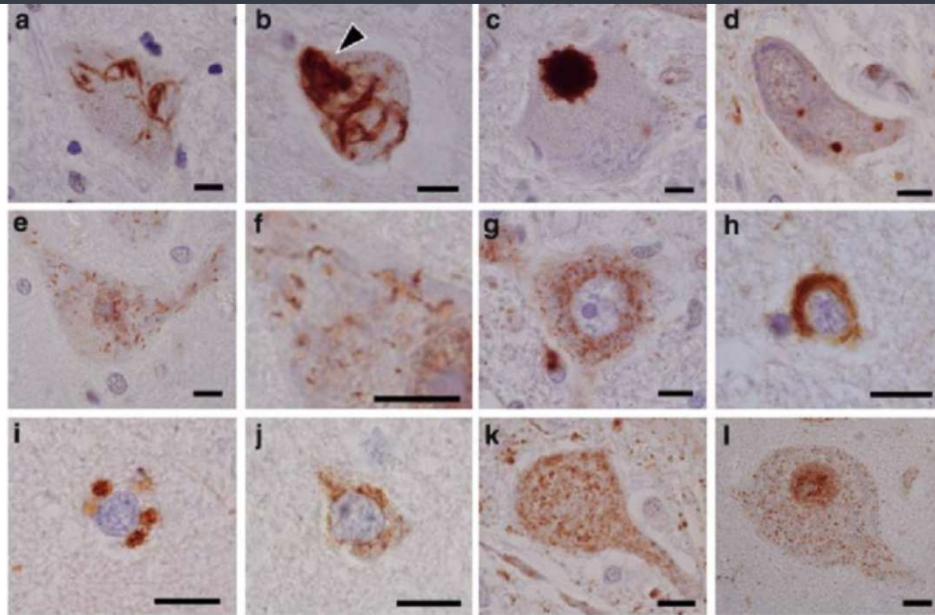


FIGURE 6-3

Bunina bodies in an anterior horn cell in the lumbar cord from a patient with amyotrophic lateral sclerosis (hematoxylin and eosin staining).

BUNINA BODIES



MATURATION OF TDP43 INCLUSIONS

Maturation process of TDP-43-positive neuronal cytoplasmic inclusions in amyotrophic lateral sclerosis with and without dementia - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/TDP-43-immunohistochemistry-in-the-brain-and-spinal-cord-of-patients-with-amyotrophic_fig1_5295566 [accessed 28 Sep, 2020]

Fig. 1 TDP-43 immunohistochemistry in the brain and spinal cord of patients with amyotrophic lateral sclerosis (ALS) (a–j) and control subjects (k, l). a and b Skein-like inclusions in the anterior horn cells. Dense compacted area (arrowhead) is also evident (b). c Rounded, spicular inclusion in the anterior horn cell. d Dot-like inclusions in the anterior horn cell. e Minute thread-like structures scattered in the cytoplasm of anterior horn cell. f Higher magnification view of the area in

GENETICS

Several Genes are implicated, and mutations are associated with both familial and sporadic disease.

SOD1 – Superoxide Dismutase: First familial form of ALS discovered, accounts for 12% of familial cases but only about 1.5% of sporadic cases.

SOD1 related ALS does not have TDP43 aggregates, poor model for disease.

SOD1 can have both AD and AR inheritance pattern

GENETICS

C9ORF72

More recently discovered

Accounts for 40% of familial cases, 7% of sporadic cases

Is associated with TDP43 aggregation

Strong association with frontotemporal dementia

Autosomal Dominant inheritance pattern

TABLE 6-6 Amyotrophic Lateral Sclerosis Genetics^a

Gene	Inheritance	Familial Amyotrophic Lateral Sclerosis ⁶²	Sporadic Amyotrophic Lateral Sclerosis ⁶²	Unique Clinical Features and Associated Phenotypes	Mechanistic Insights ⁶³
<i>SOD1</i>	Autosomal dominant, autosomal recessive	12%	1.5%	Lower motor neuron signs prominent	Mechanism likely involves oxidative stress
<i>TARDBP</i>	Autosomal dominant	4%	1%	Onset usually in upper limbs	Ribonucleic acid (RNA) processing
<i>FUS</i>	Autosomal dominant, autosomal recessive	4%	1%	Rare frontotemporal dementia (FTD)	RNA processing
<i>VCP</i>	Autosomal dominant	1%	1%	FTD, Paget disease, inclusion body myositis, parkinsonism	Protein degradation
<i>C9ORF72</i>	Autosomal dominant	40%	7%	FTD	RNA processing, membrane trafficking
<i>SQSTM1</i>	Autosomal dominant	1%	<1%	Paget disease, FTD, inclusion body myositis	Protein degradation
<i>OPTN</i>	Autosomal dominant, autosomal recessive	<1%	<1%	Open-angle glaucoma, Paget disease	Trafficking, protein degradation
<i>PFN1</i>	Autosomal dominant	<1%	<1%	None described	Cytoskeletal processing
<i>UBQLN2</i>	X-linked dominant	<1%	<1%	Juvenile, upper motor neuron signs prominent, FTD	Protein degradation

^a Modified from Goutman S, Handbook of Clinical Neurology.⁶¹ © 2017 Elsevier.

TABLE OF ASSOCIATED GENETIC DISEASE IN ALS

ALS - CLINICAL MANIFESTATIONS

- PAINLESS PROGRESSIVE WEAKNESS
- THE TYPICAL FORM (SPINAL FORM) MANIFESTS AS INSIDIOUS ONSET OF **WEAKNESS IN A SINGLE DISTAL LIMB**
- ROSTRAL -> CAUDAL AND CONTRALATERAL SPREAD
- **BULBAR ONSET** CAN OCCUR IN 20-30% OF PATIENTS
- MORE RARELY, ONSET WITH RESPIRATORY WEAKNESS OR FRONTOTEMPORAL DEMENTIA
- RELENTLESS PROGRESSION IN MOST
- PROGRESS TO THE POINT THAT RESPIRATION AND SECRETIONS CANNOT BE MANAGED WITHOUT TRACHEOSTOMY
- SOME ELECT TRACHEOSTOMY AND G-TUBE
- IMPORTANT TO RECOGNIZE THAT THIS IS **NOT A PURELY MOTOR DISEASE**
- UP TO 15% WILL DEVELOP FRONTOTEMPORAL DEMENTIA DURING ILLNESS
- UP TO 50% WILL DEVELOP MEASURABLE EXECUTIVE DYSFUNCTION

TABLE 6-1 Amyotrophic Lateral Sclerosis Phenotypes^a

Clinical Phenotype	Segmental Involvement			Comments
	Upper Motor Neuron	Lower Motor Neuron	Other	
Classic amyotrophic lateral sclerosis (ALS)	Bulbar, upper extremity, lower extremity	Bulbar, upper extremity, lower extremity	Cognitive impairment, frontotemporal dementia	Can begin in limbs or bulbar region
Progressive bulbar palsy	Bulbar	Bulbar	Pseudobulbar affect sometimes present	May remain isolated; more common in females
Respiratory-onset ALS	May develop in bulbar, upper extremity, lower extremity	May develop in bulbar, upper extremity, lower extremity		Early diaphragmatic involvement and neuromuscular respiratory failure
Progressive muscular atrophy	May develop clinically or be seen at autopsy; see article text	Lower extremity	Cognitive impairment	Weakness often proximal but can be distal
Primary lateral sclerosis	Bulbar, upper extremity, lower extremity	Discrete involvement	Cognitive impairment	Asymmetric
Pseudopolyneuritic ALS		Upper extremity, lower extremity		Weakness in distal limbs
Hemiplegic ALS	Upper extremity, lower extremity	Discrete involvement		
Flail arm syndrome (brachial amyotrophic diplegia)	Lower extremity	Upper extremity		Symmetric and proximal predominance; bulbar symptoms can develop later in disease course
Flail leg syndrome		Lower extremity		Weakness can be symmetric or asymmetric and often distal onset
Dropped head syndrome			Neck weakness	Exclude neuromuscular junction disease and myopathy
ALS frontotemporal dementia	Bulbar, upper extremity, lower extremity	Bulbar, upper extremity, lower extremity	Frontotemporal dementia	Dementia can occur before ALS symptoms

^a Data from Swinnen B, Robberecht W. *Nat Rev Neurol*.² nature.com/nrnjournal/v10/n11/abs/nrneurol.2014.184.html and Al-Chalabi A, Hardiman O. *Nat Rev Neurol*.³ nature.com/nrnjournal/v9/n11/abs/nrneurol.2013.203.html.

ALS PHENOTYPES

DIAGNOSIS

- LOWER MOTOR NEURON FINDINGS + UPPER MOTOR NEURON FINDINGS
- LMN FINDINGS:
 - WEAKNESS
 - FASCICULATIONS
 - ATROPHY
- UMN FINDINGS:
 - SPASTICITY
 - HYPERREFLEXIA
 - SPREADING REFLEXES
 - PATHOLOGICAL REFLEXES (EX UP-GOING BABINSKI)
- **REVISED EL ESCORIAL CRITERIA ARE COMMONLY USED**
- DIAGNOSIS IS FREQUENTLY DELAYED BY UP TO A YEAR

TABLE 6-4 Comparison of El Escorial, Revised El Escorial, and Awaji Criteria

Diagnostic categories ^a	Criteria		
	El Escorial ²²	Revised El Escorial ⁵	Awaji ²³
Clinically definite	Clinical evidence of: (1) UMN plus LMN signs in the bulbar and two spinal regions OR (2) UMN plus LMN signs in three spinal regions	Same as El Escorial	LMN signs are defined by clinical or electrophysiologic evidence; otherwise same
Clinically probable	Clinical evidence of: UMN plus LMN signs in at least two regions with UMN signs rostral to LMN signs	Same as El Escorial	LMN signs are defined by clinical or electrophysiologic evidence; otherwise same
Clinically probable—laboratory-supported	Not included	Clinical evidence of: (1) UMN plus LMN signs in one region OR UMN signs alone in one region AND (2) LMN by EMG criteria in at least two regions	Not included
Clinically possible	Clinical evidence of: (1) UMN plus LMN signs in one region OR (2) UMN signs in two or more regions OR (3) LMN signs are rostral to UMN signs	Same as El Escorial	LMN signs are defined by clinical or electrophysiologic evidence; otherwise same
Clinically suspected	Clinical evidence of: LMN signs in two or more regions	Diagnostic category deleted in the revised El Escorial criteria	

Continued on page 1342

TABLE 6-4 Comparison of El Escorial, Revised El Escorial, and Awaji Criteria *Continued from page 1341*

Electrophysiologic features ^b	Criteria		
	El Escorial ²²	Revised El Escorial ⁵	Awaji ²³
Active denervation		Fibrillation potentials or positive sharp waves	Same; additionally, fasciculation potentials in muscles with chronic denervation
Chronic denervation		Large motor unit potentials, reduced interference pattern with firing rate greater than 10 Hz, or unstable motor unit potentials	Same
Sensitivity and specificity			
Sensitivity ²⁴		45%	57%
Specificity ²⁴		99.5%	99.5%

EMG = electromyography; LMN = lower motor neuron; UMN = upper motor neuron.

^a Neuroimaging and clinical tests exclude other diagnoses.^b As defined by the El Escorial criteria, electrophysiologic evidence of LMN degeneration in at least two muscles of different spinal root and peripheral nerve innervation in two or more regions confirms an amyotrophic lateral sclerosis diagnosis.

REVISED EL ESCORIAL CRITERIA

Revised el Escorial allows for a laboratory supported clinically probable diagnosis of ALS. In other words a diagnosis can be made by substituting a second region's involvement for EMG findings (allows for earlier diagnosis)

ELECTRODIAGNOSTIC STUDIES

- PERFORMED TO CONFIRM LOWER MOTOR DYSFUNCTION AND RULE OUT OTHER PROCESSES (SUCH AS A NEUROPATHY)
- EMG SHOWS POLYPHASIC MOTOR UNIT ACTION POTENTIALS – OCCURS WHEN MUSCLE FIBERS ARE REINNERVATED
- RECALL THAT SURVIVING ALPHA MOTOR NEURONS CAN REINNERVATE MUSCLE UNITS THAT HAVE BEEN DENERVATED
- **CANNOT ESTABLISH UPPER MOTOR NEURON FINDINGS (ONLY CLINICAL EXAM CAN DO THIS)**
- FIBRILLATION POTENTIALS AND FASCICULATION POTENTIALS ARE SEEN
- EMG CAN BECOME POSITIVE IN A MUSCLE BEFORE CLINICAL WEAKNESS IS DETECTED, WHICH CAN AID IN ESTABLISHING A DIAGNOSIS EARLIER

LABORATORY STUDIES

- NO GUIDELINE CONSENSUS ON LABS
- AT A MINIMUM:
 - COMPREHENSIVE METABOLIC PANEL
 - CREATINE KINASE (TO EVALUATE FOR MYOPATHY WHICH SHOULD NOT BE PRESENT IN ALS)
 - B12
 - COPPER
 - SERUM AND URINE ELECTROPHORESIS

TABLE 6-2 Diagnostic Evaluation for Amyotrophic Lateral Sclerosis^a

Test	Diagnostic Considerations	Commonly Performed
Clinical chemistry		
General chemistry (electrolytes, liver function tests) ^b	Hypercalcemia, hypocalcemia, hypophosphatemia, hepatic myelopathy	Yes
Endocrine studies ^c	Hyperthyroidism, thyroid myopathy, hyperparathyroidism, Allgrove syndrome, diabetic amyotrophy, insulinoma	No
Creatine kinase ^d	Myopathies	Yes
Vitamin B ₁₂	Subacute combined degeneration	Yes
Copper	Copper deficiency myelopathy	Yes
Heavy metals	Heavy metal poisoning	No
Inflammatory/autoimmune (ESR, CRP, ANA, ANCA, RF)	Inflammatory/autoimmune diseases	No
Serum protein electrophoresis, immunofixation electrophoresis ^e	Motor neuropathy with a monoclonal gammopathy	Yes
Antiganglioside antibodies	Multifocal motor neuropathy, dysimmune lower motor neuron syndromes (with GM1, GD1b, and asialo-GM1 antibodies)	No
Infectious studies (syphilis, Lyme disease, HIV, HTLV) ^f	Syphilitic meningomyelitis, neuroborreliosis, Bannwarth syndrome, HIV-related motor neuropathy, HTLV-associated ALS-like syndrome, poliomyelitis, West Nile virus	No
Hexosaminidase	Late-onset Tay-Sachs disease	No
Anti-Hu ^g	Paraneoplastic disorders	No
Anti-MAG	Anti-MAG peripheral neuropathy	No
Anti-acetylcholine receptor and anti-muscle-specific kinase antibodies	Myasthenia gravis	No
CSF		
Cell count		No
Protein concentration		No
Cytology		No
IgG index		No
Serology (<i>Borrelia burgdorferi</i> , virus)		No
Urine		
Heavy metals		No
Urine immunofixation		No

FURTHER
DIAGNOSTIC
TESTS TO BE
CONSIDERED

ALS TREATMENTS

QUALITY OF LIFE

- CURRENTLY NO CURE
- FOCUS IS ON SYMPTOMATIC MANAGEMENT, QUALITY OF LIFE, AND GOALS OF CARE
- ALS PATIENTS SHOULD BE REFERRED TO AN ALS MULTIDISCIPLINARY CLINIC
 - TEAMS USUALLY INCLUDE A NEUROLOGIST AND/OR PULMONOLOGIST, PHYSICAL AND OCCUPATION THERAPISTS, NUTRITIONIST, SPEECH AND LANGUAGE THERAPY.
- **THE IMPORTANCE OF GOALS OF CARE DISCUSSIONS CANNOT BE OVERSTATED**

MEDICATIONS

Riluzole

- Shown to confer a 2-3 month survival benefit in clinical trials
- Potentially more benefit in those with bulbar onset and younger patients
- Relatively good safety profile, though long-term use may be associated with hepatic toxicity.

Edaravone

- Free radical scavenger – thought to reduce oxidative stress
- Phase 2 trials showed safety and reduction in CSF biomarkers for oxidative stress
- Phase 3 trial failed to show benefit in primary end points of functional improvement
- Second Phase 3 trial with more selective inclusion criteria showed a benefit in functional decline over 2 years.
- IV infusion

Priority number one is to determine if invasive mechanical ventilation is consistent with the patient's wishes.

Priority number one is to determine if invasive mechanical ventilation is consistent with the patient's wishes.

Initial manifestations usually occur during sleep

Non-invasive ventilation confers a survival benefit, greater in those with bulbar onset

Usually initiated when FVC < 50% of expected or maximum inspiratory pressure drops below 60 cm H₂O

Some patient's will elect for tracheostomy when non-invasive ventilation fails

MECHANICAL VENTILATION

FEEDING

Bulbar weakness leads to dysphagia

Dysphagia increases risk for aspiration, which in an already weak respiratory system can easily be fatal

Gastrostomy tube is an option for feeding if dysphagia is bad enough, though like tracheostomy needs to be clearly in line with their wishes

ALS MIMICS

BE AWARE THAT THERE
ARE SEVERAL DISEASES
THAT MUST BE EXCLUDED
BEFORE A DIAGNOSIS OF
ALS IS MADE

ALS MIMICS

Diagnosis	Diagnostic Clue	Confirmatory Testing
Lower motor neuron predominant/weakness		
Benign fasciculations	Acute onset, widespread, no weakness	EMG does not demonstrate denervation or chronic reinnervation (and often does not show fasciculations)
Inclusion body myopathy	Weakness of deep finger flexors and quadriceps	Myopathic EMG, plus NTSC1A antibody (helpful when present), muscle biopsy findings
Multifocal motor neuropathy (MMN) with conduction block	Nerve (rather than myotome) pattern with asymmetric upper extremity predominance	Partial motor conduction block on nerve conduction studies, positive anti-GM1 antibodies
Neuralgic amyotrophy	Pain at onset, involvement of named nerves, self-limited course	Nerve conduction studies and EMG findings, MRI with and without contrast of the involved plexus
Monomelic amyotrophy (Hirayama disease)	Young male with asymmetric hand and distal forearm weakness and atrophy, self-limited course	MRI findings
Spinal bulbar muscular atrophy (Kennedy disease)	Slow progression, facial twitching, tremor, sensory neuropathy, evidence of androgen insensitivity (eg, gynecomastia, testicular atrophy)	Absent sural sensory responses, CAG repeat expansion in the androgen receptor gene
Motor-predominant Charcot-Marie-Tooth disease (CMT)/distal spinal muscular atrophy	Symmetric and distal onset, young age of onset, slow progression	Nerve conduction studies demonstrating abnormal sensory nerve action potential (SNAP) amplitudes and (in CMT type 1) evidence of demyelination; positive genetic testing for CMT
Post-severe denervation (postpolio) syndrome	History of distant polio or other severe nerve injury with recovery followed by slow progression of weakness in the distribution of prior polio symptoms; muscle pain is common	Giant motor unit action potentials on EMG

SPINOBULBAR MUSCULAR ATROPHY

- ALSO CALLED KENNEDY DISEASE
- PROGRESSIVE DISORDER OF ANTERIOR HORN CELLS
- CAUSES PROXIMAL LIMB WEAKNESS AND BULBAR WEAKNESS
- X-LINKED RECESSIVE DISEASE, CAG REPEAT EXPANSION ON ANDROGEN RECEPTOR GENE
- ASSOCIATED WITH ENDOCRINE ABNORMALITIES SUCH AS GYNECOMASTIA, DM, INFERTILITY
- CK ELEVATED > 1000 IU/L
- DIAGNOSED BY GENETIC EVALUATION
- SURVIVAL 2 – 3 DECADES AFTER ONSET

SPINAL MUSCULAR ATROPHY

TABLE 6-7 Spinal Muscular Atrophy Phenotypes^a

Type	Onset Age	Best Motor	Survival	Cases
0	Prenatal	Respiratory failure at birth	Weeks	<5%
1 (Werdnig-Hoffmann disease)	Birth to 6 months	Unable to sit	Before age 1 to 2	45%
2	Before 18 months	Unable to stand or walk	Early adulthood	20%
3 (Kugelberg-Welander disease)	After 18 months	Able to stand and walk	Adulthood	30%
4	Adulthood	Ambulatory as adult	Adulthood	<5%

^a Modified with permission from Arnold WD, et al, Muscle Nerve.⁸¹ © 2014 Wiley Periodicals, Inc. onlinelibrary.wiley.com/doi/10.1002/mus.24497/full.

- 4 TYPES, CAUSE PROGRESSIVE MUSCULAR WEAKNESS DUE TO LOWER MOTOR NEURON DEGENERATION MUTATION OF SMN GENE
- TWO FORMS OF SMN GENE EXIST IN HUMANS, SMN1 AND SMN2
- SMN2 GENE IS ABLE TO FORM A FUNCTIONAL SMN PROTEIN IN ABOUT 10% OF TRANSCRIPTS
- SMN1 DEFICIENCY CAUSES THE DISEASE, BUT SMN2 COPY NUMBER IS THOUGHT TO ACCOUNT FOR SOME OF THE WIDE SPECTRUM OF DISEASE
- **NUSINERSIN, A FIRST-IN-CLASS MEDICATION WAS RECENTLY APPROVED FOR TREATMENT**
 - ANTISENSE OLIGONUCLEOTIDE WHICH PROMOTES SMN2 TRANSCRIPTION, THEREBY INCREASING THE AMOUNT OF FUNCTIONAL SMN PROTEIN

SPINAL MUSCULAR ATROPHY

TABLE 6-7 Spinal Muscular Atrophy Phenotypes^a

Type	Onset Age	Best Motor	Survival	Cases
0	Prenatal	Respiratory failure at birth	Weeks	<5%
1 (Werdnig-Hoffmann disease)	Birth to 6 months	Unable to sit	Before age 1 to 2	45%
2	Before 18 months	Unable to stand or walk	Early adulthood	20%
3 (Kugelberg-Welander disease)	After 18 months	Able to stand and walk	Adulthood	30%
4	Adulthood	Ambulatory as adult	Adulthood	<5%

^a Modified with permission from Arnold WD, et al, Muscle Nerve.⁸¹ © 2014 Wiley Periodicals, Inc. onlinelibrary.wiley.com/doi/10.1002/mus.24497/full.

- ONASEMNOGENE ABEPARVOVEC (ZOLGENSMA)
- ONE-TIME IV INFUSION OF AN ADENOVIRUS VECTOR CONTAINING SMN1 TRANSGENE + PROMOTORS
- INCREASES SMN PROTEIN
- IN A SMALL TRIAL, INCREASE SURVIVAL FROM 8% TO 100%.

MYASTHENIA SUMMARY:

Disorder of impaired neuromuscular junction transmission

Autoimmune – in most cases caused anti-Ach receptor antibodies

Variable presentation but all include easy fatiguability, often worsening throughout the day

Diagnosis is clinical + serology +/- emg/ncs

Tx with pyridostigmine + steroids +/- other immunologics and biologics

Always check for thymoma with non-contrast CT Chest

Thymectomy for thymomatous MG

Thymectomy for nonthymomatous MG too if age 18-65

ALS SUMMARY

Neurodegenerative disorder of motor neurons, in classic form affects both upper and lower

Clinical + Electrophys diagnosis (UMN signs by exam only)

Numerous genetic syndromes have been discovered, be familiar with SOD1, FUS, and C9orf72

No cure, riluzole and edaravone may help to slow functional decline

Treatment focus is on quality of life. Goals of care discussions are of prime importance

Patients should be referred to a multidisciplinary specialty clinic